

Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Review Calculus

1. Differentiation

a. Find the equation of the tangent line to the curve $y = 2x^3 - x^2 + 3$ at the point where $x = 1$.

b. Determine the critical points and their concavity of the function $f(t) = te^{(-t^2)}$.

2. Integration

a. Integrate $\int x (\ln (x))^2 dx$ using integration by parts.

b. Use integration by parts and limits to show that $\int_1^\infty \frac{\ln (t)}{t^2} dt = 1$ is true.